

Credit Delinquency Prediction Model – Submission

Step 1: Model Logic

Predictive Modeling Approach:

A supervised machine learning pipeline using Logistic Regression was designed to predict credit delinquency.

Model Options:

- Logistic Regression (Simple): Interpretable and suitable for binary classification
- Random Forest (Complex): Higher accuracy but less interpretable

Selected Model: Logistic Regression

Key Features:

Credit_Utilization, Debt_to_Income_Ratio, Missed_Payments, Total_Missed_Payments, Credit_Score

Model Workflow:

1. Data ingestion
2. Feature selection
3. Data preprocessing
4. Train-test split
5. Model training
6. Prediction
7. Evaluation

Final Model Summary:

Logistic Regression predicts delinquency probability using financial and behavioral features, offering interpretability and suitability for financial decision-making.

Step 2: Model Justification

Logistic Regression is appropriate due to its high interpretability, essential in financial services. It explains how variables influence risk, ensuring transparency and compliance.

Although Random Forest may improve accuracy, it lacks explainability. Logistic Regression is efficient, scalable, and suitable for real-time risk assessment.

Step 3: Evaluation Strategy

Performance Metrics:

Accuracy, Precision, Recall, F1 Score, AUC-ROC

Reliability Checks:

Cross-validation, Train vs test comparison

Fairness Checks:

Evaluate across income, employment, and location groups

Bias Mitigation:

Remove sensitive variables, balance dataset, monitor fairness

Interpretation:

High Recall captures risky customers

High Precision reduces false alarms

AUC > 0.75 indicates strong performance